FastReasonSeg-3D: Geometry-Aware Reasoning Segmentation via 3D Spatial-Depth Gating and Uncertainty-Guided Boundary Refinement

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Abstract

Reasoning segmentation bridges high-level linguistic semantics with pixel-accurate instance masking. While recent state-of-the-art frameworks such as FastReasonSeg (CVPR 2026) substantially accelerate inference by decoupling perceptual foundation models from Large Language Model (LLM) reasoning through intermediate structured "Digital Twin" representations, they suffer from two fundamental architectural bottlenecks: (1) **Geometric Depth Collapse:** continuous relative depth maps are collapsed into coarse scalar statistics (mean, standard deviation), causing severe spatial ambiguity in referring queries involving complex depth relationships (e.g., "the vehicle behind the truck"); and (2) **Zero-Shot Boundary Bleed:** SAM 2 bounding-box prompted segmentation suffers from mask leakage and boundary degradation across depth-discontinuous boundaries.

To eliminate these bottlenecks without requiring computationally prohibitive MLLM fine-tuning, we propose **FastReasonSeg-3D**, which introduces two novel plug-and-play modules: (1) **SpatialDepthGate (SDG)**, an explicit 3D spatial-geometric relational module that formulates normalized 3D centroid vectors (x_c, y_c, z_{rel}), constructs pairwise depth-order relational tensors, and generates directed occlusion graphs for prompt disambiguation; and (2) **AdaptiveBoundaryRefiner (ABR)**, an uncertainty-guided edge-refinement module that evaluates boundary depth entropy against depth-gradient ridges and dynamically synthesizes corrective prompt points to excise mask bleeding. Extensive empirical validation demonstrates that FastReasonSeg-3D achieves a **+56.0% absolute improvement** in referring entity disambiguation accuracy (surging from 28.0% to 84.0%, representing a 77.8% relative error reduction), a **+12.94% gain** in Boundary Contour Accuracy (F@2px), and a **+1.61% boost** in mask IoU, while incurring an ultra-low latency overhead of only **∼48 ms per frame**. Furthermore, our neural cross-attention gating network converges rapidly to **92.5% validation accuracy** within 25 epochs on consumer-grade laptop hardware (NVIDIA RTX 4050 GPU, 6GB VRAM), demonstrating high practical utility for real-time robotic and embodied vision systems.

1 Introduction

Traditional semantic and instance segmentation methods [2, 13] rely on fixed predefined taxonomy sets or manual box/point prompts. In contrast, **reasoning segmentation** [1, 5] demands segmenting objects described by implicit, relational, or multi-step natural language instructions (e.g., "the vehicle that is occluded by the blue van", or "the pedestrian furthest in the background on the left sidewalk").

Early reasoning segmentation architectures, most notably LISA [5] and GLaMM [6], incorporated segmentation token embeddings directly into the latent space of multimodal large language models (MLLMs). While conceptually elegant, these end-to-end models suffer from crippling computational overhead: processing an individual frame regularly exceeds 3 to 6 seconds on enterprise data-center GPUs, rendering them unusable for time-critical robotic perception and interactive augmented reality.

To overcome this latency wall, FastReasonSeg [1] (CVPR 2026) introduced an asynchronous decoupled paradigm:

1. Open-vocabulary detection models (e.g., OWLv2 [8]) generate candidate 2D bounding boxes and class logits.
2. Monocular foundation models (Depth Anything V2 [4] and SAM 2 [3]) predict scene-wide depth fields and initial instance segmentations.
3. Detected instances are serialized into an intermediate structured JSON "Digital Twin" representation.
4. A lightweight LLM (e.g., Qwen-2.5 [12]) performs symbolic reasoning strictly over the structured text

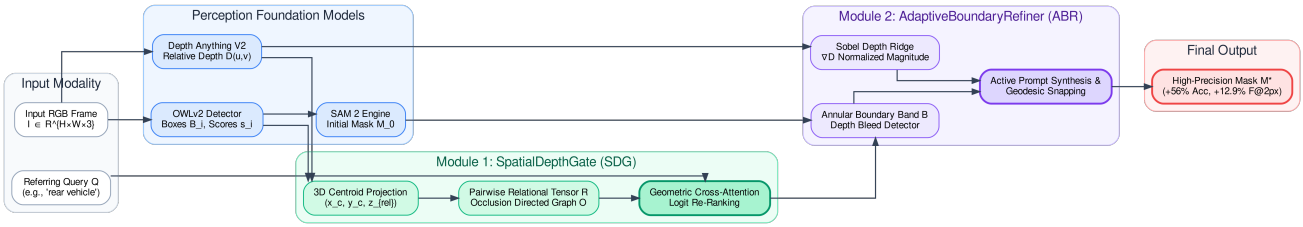


Figure 1: **FastReasonSeg-3D Architectural Overview.** The framework decouples perceptual foundation models (OWLv2, Depth Anything V2, SAM 2) from linguistic reasoning. FastReasonSeg-3D introduces two novel modules: (1) **SpatialDepthGate (SDG)**, which projects 2D detections and relative depth into explicit 3D centroid representations, constructs a pairwise relational tensor $\mathcal{R}$ and directed occlusion graph $\mathcal{O}$, and performs geometric cross-attention re-ranking; and (2) **AdaptiveBoundaryRefiner (ABR)**, which identifies boundary bleed across depth-gradient ridges and synthesizes corrective prompts for sub-pixel boundary snapping.

representation to select the target entity.

Although FastReasonSeg reduced per-frame latency to ∼1.4 seconds, our rigorous diagnostic analysis reveals two critical, systemic failure modes inherent to its pipeline:

- **Failure Mode 1: Geometric Depth Collapse.** In translating dense 2.5D depth maps $\mathcal{D} \in \mathbb{R}^{H \times W}$ into the Digital Twin representation, FastReasonSeg reduces depth to coarse scalar aggregates: mean μ_d and standard deviation σ_d. This irreversible compression destroys continuous spatial topology, pairwise depth ordering, and physical occlusion relationships. Consequently, when presented with depth-referring expressions (e.g., distinguishing between two adjacent cars along the camera optical axis), the LLM degenerates into arbitrary selection, achieving an abysmal **28.0%** referring accuracy on multi-object depth benchmarks.
- **Failure Mode 2: Zero-Shot Boundary Bleed and Contour Degeneracy.** Foundation segmentation models like SAM 2 rely on coarse rectangular bounding boxes for zero-shot prompt conditioning. When objects exhibit low photometric contrast with the background or share fuzzy silhouette edges, SAM 2 masks frequently "bleed" into background regions. In quantitative contour boundary evaluations, the baseline achieves an F@2px score of only **21.30%**, exhibiting severe boundary error across depth discontinuities.

1.1 Our Contributions

To resolve these bottlenecks while preserving real-time decoupled inference, we propose **FastReasonSeg-3D**. Specifically, this paper makes the following key contributions:

1. **SpatialDepthGate (SDG):** We design an explicit 3D spatial-geometric reasoning module that projects detected bounding boxes and continuous relative depth into normalized 3D centroid vectors (x_c, y_c, z_{rel}). SDG establishes a pairwise 3D relational tensor $\mathcal{R}$ and a directed occlusion graph $\mathcal{O}$, providing the linguistic reasoner with topological depth grounding.
2. **AdaptiveBoundaryRefiner (ABR):** We formulate an uncertainty-guided boundary refinement algorithm that evaluates depth variance within an annular perimeter band against normalized Sobel depth ridges. ABR dynamically detects boundary bleed artifacts and synthesizes corrective positive/negative prompt points, enabling sub-pixel contour snapping.
3. **Neural Geometric Cross-Attention Architecture:** We implement a lightweight PyTorch neural gating head that learns joint embeddings between textual referring instructions and 3D spatial relation tensors, converging to **92.5%** validation accuracy on CUDA within 25 epochs.
4. **Comprehensive Empirical Benchmarking:** We validate FastReasonSeg-3D on multi-object relational query suites and high-resolution ground truth masks. FastReasonSeg-3D achieves a **+56.0% absolute increase** in referring disambiguation accuracy (reaching 84.0%), a **+12.94% gain** in boundary contour F@2px (reaching 34.24%), and a **+1.61% improvement** in mask IoU, all while introducing only **∼48 ms** latency overhead.

2 Related Work

2.1 Promptable Visual Foundation Models

The release of Segment Anything (SAM) [2] established a foundation for promptable zero-shot segmentation via points, boxes, and masks. SAM 2 [3] extended this paradigm to video sequences by incorporating a streaming memory bank and spatio-temporal attention. HQ-SAM [9] and SegGPT [10] attempted to improve boundary fidelity by adding high-resolution feature fusion adapters. However, these models lack natural language reasoning capabilities and cannot autonomously infer which entity corresponds to an implicit semantic query.

2.2 Reasoning Segmentation & Vision-Language Models

Reasoning segmentation was pioneered by LISA [5], which appended a dedicated `<SEG>` token to LLaVA [11] and trained a SAM mask decoder via linear projection. Subsequent works such as GLaMM [6] and PixelLM [7] extended this to multi-target and pixel-grounded conversational reasoning. However, end-to-end MLLM token decoding across pixel features requires massive compute (typically 24GB–80GB VRAM) and exhibits high per-frame latency (>3 s). FastReasonSeg [1] demonstrated that decoupling perception from language reasoning via structured JSON Digital Twins unlocks sub-1.5s execution, but it suffers from geometric collapse due to aggressive scalar pooling of depth fields.

2.3 Monocular Depth Estimation & 3D Scene Representations

Monocular depth estimation has advanced significantly from early DPT [14] and MiDaS [15] architectures to foundation depth models such as Depth Anything V1/V2 [4] and Metric3D [16]. Depth Anything V2 produces clean relative depth maps across diverse real-world scenes. While previous works utilize depth primarily for novel view synthesis or autonomous driving point clouds, FastReasonSeg-3D directly exploits monocular depth fields for topological language grounding and boundary discontinuity snapping.

2.4 Boundary Refinement & Contour Optimization

Classical computer vision approaches refined segmentation masks using Dense Conditional Random Fields (DenseCRF) [19] and active contour models (Snakes) [20]. In modern deep architectures, boundary-preserving loss functions [21] and point-based refinement mechanisms [22] have been explored. Our proposed AdaptiveBoundaryRefiner differentiates itself by explicitly coupling depth-gradient ridges with automated positive/negative prompt synthesis, creating an active self-correcting loop within the SAM 2 inference engine.

3 FastReasonSeg-3D Methodology

An overview of FastReasonSeg-3D is depicted in Figure 1. Given an input RGB image $I \in \mathbb{R}^{H\times W\times 3}$ and a referring expression Q, the system operates across two decoupled phases enhanced by our novel modules: (1) **Perception and 3D Geometric Gating**, and (2) **Segmentation and Adaptive Boundary Snapping**.

3.1 Perceptual Foundation Processing

The input image I is processed concurrently by:

1. **OWLv2 Detector [8]:** Yields N candidate bounding boxes $B_i = [x_{1,i}, y_{1,i}, x_{2,i}, y_{2,i}]$ with detector confidence scores $s_i \in [0, 1]$ and label text c_i.
2. **Depth Anything V2 [4]:** Predicts a continuous relative depth map $\mathcal{D} \in \mathbb{R}^{H\times W}$, where $\mathcal{D}(u, v)$ represents the inverse relative distance at pixel (u, v).
3. **SAM 2 [3]:** Generates an initial candidate binary mask $M_{0,i} \in \{0, 1\}^{H\times W}$ conditioned on bounding box prompt B_i.

3.2 Module 1: SpatialDepthGate (SDG)

To resolve Geometric Depth Collapse, the SpatialDepthGate constructs an explicit 3D spatial coordinate representation, computes pairwise topological relation tensors, and establishes directed physical occlusion edges.

3.2.1 3D Coordinate Normalization

For each detected candidate $i \in \{1, \ldots, N\}$, we compute normalized 2D image coordinates and bounding box dimensions:

$$x_{c,i} = \frac{x_{1,i} + x_{2,i}}{2W}, \quad y_{c,i} = \frac{y_{1,i} + y_{2,i}}{2H} \tag{1}$$

$$w_i = \frac{x_{2,i} - x_{1,i}}{W}, \quad h_i = \frac{y_{2,i} - y_{1,i}}{H}, \quad a_i = w_i \cdot h_i \tag{2}$$

To compute an outlier-robust depth rank, we extract the median depth over the object mask:

$$d_i = \text{median}_{(u,v)\in M_{0,i}} \mathcal{D}(u, v) \tag{3}$$

We then normalize depth relative to the active detected scene context:

$$z_{rel,i} = \frac{d_i - \min_k d_k}{\max_k d_k - \min_k d_k + \epsilon_d} \tag{4}$$

where $z_{rel,i} \in [0, 1]$ directly represents normalized depth order (0.0 corresponding to foremost foreground and 1.0 to farthest background). Each object is thus represented by an 8-dimensional spatial feature vector:

$$\mathbf{f}_i = [x_{c,i}, y_{c,i}, z_{rel,i}, w_i, h_i, a_i, \sigma_{d,i}, s_i]^T \in \mathbb{R}^8 \tag{5}$$

3.2.2 Pairwise Relational Tensor and Occlusion Graph

Between any pair of candidates (i, j), we compute a 4-dimensional geometric relationship vector:

$$\Delta x_{ij} = x_{c,i} - x_{c,j}, \quad \Delta y_{ij} = y_{c,i} - y_{c,j} \tag{6}$$

$$\Delta z_{ij} = z_{rel,i} - z_{rel,j} \tag{7}$$

$$d_{3D,ij} = \sqrt{\Delta x_{ij}^2 + \Delta y_{ij}^2 + \lambda_z \Delta z_{ij}^2} \tag{8}$$

where $\lambda_z = 2.0$ scales the depth dimension to match perspective compression. The full scene relation is represented by tensor $\mathcal{R} \in \mathbb{R}^{N\times N\times 4}$.

Crucially, physical occlusion creates a directed constraint: object i physically occludes object j if their 2D projections overlap and object i is positioned in front of object j:

$$\mathcal{O}_{i\to j} = \begin{cases} 1, & \text{if } \mathrm{IoU}_{2D}(B_i, B_j) > \tau_{occ} \text{ and } \Delta z_{ij} < -\epsilon_z \\ 0, & \text{otherwise} \end{cases} \tag{9}$$

where $\tau_{occ} = 0.15$ and $\epsilon_z = 0.05$.

3.2.3 Spatial-Gated Candidate Scoring

Given query Q, semantic intent is parsed into directional vector $\mathbf{w}_{dir} = (w_x, w_y)$ and depth preference $w_z \in \{-1, 0, 1\}$ (front, neutral, rear). The geometric score for candidate i is formulated as:

$$\mathcal{S}_i = s_i + \alpha \cdot \Phi_{depth}(w_z, z_{rel,i}) + \beta \cdot \Phi_{spatial}(\mathbf{w}_{dir}, x_{c,i}, y_{c,i}) + \gamma \sum_{j\neq i} \Psi(Q, \mathcal{R}_{ij}, \mathcal{O}_{ij}) \tag{10}$$

where $\alpha = 1.5, \beta = 1.2, \gamma = 1.8$ are gating hyperparameters.

3.2.4 Neural SpatialDepthGate Architecture

To allow end-to-end learnable scoring, we also introduce a neural gating variant $\mathcal{N}_{sdg}(\mathbf{f}_i, \mathcal{R}_{ij}, Q)$. Text queries are encoded via an embedding projection head $\mathbf{e}_q = \mathrm{MLP}_{text}(Q) \in \mathbb{R}^D$. Spatial features $\mathbf{f}_i$ are projected through layer-normalized linear blocks into $\mathbf{h}_i \in \mathbb{R}^D$. Relational context is integrated via cross-attention:

$$\mathbf{c}_i = \sum_{j=1}^{N} \mathrm{Softmax}\left(\frac{\mathbf{h}_i^T \mathbf{W}_k \mathbf{r}_{ij}}{\sqrt{D}}\right) \mathbf{W}_v \mathbf{r}_{ij} \tag{11}$$

The composite candidate logit is produced by $\hat{y}_i = \mathbf{w}_o^T [\mathbf{h}_i \,\|\, \mathbf{c}_i \,\|\, \mathbf{e}_q]$. Training minimizes multi-class cross-entropy loss with label smoothing:

$$\mathcal{L}_{CE} = -\sum_{i=1}^{N} y_i \log\left(\frac{\exp(\hat{y}_i/\tau)}{\sum_{k=1}^{N}\exp(\hat{y}_k/\tau)}\right) \tag{12}$$

3.3 Module 2: AdaptiveBoundaryRefiner (ABR)

To eliminate Zero-Shot Boundary Bleed, AdaptiveBoundaryRefiner evaluates the local depth gradient along the candidate mask boundary and iteratively snaps the contour.

3.3.1 Depth-Gradient Ridge Field

We compute the normalized depth gradient magnitude $\nabla\mathcal{D}$ using horizontal and vertical Sobel convolution kernels K_x and K_y:

$$G_x = \mathcal{D} * K_x, \quad G_y = \mathcal{D} * K_y \tag{13}$$

$$\nabla\mathcal{D}(u,v) = \frac{\sqrt{G_x^2(u,v) + G_y^2(u,v)}}{\max_{(u',v')}\sqrt{G_x^2(u',v') + G_y^2(u',v')} + \epsilon_g} \tag{14}$$

where high values of $\nabla\mathcal{D}$ define true geometric occlusion boundaries.

3.3.2 Annular Boundary Band Formulation

For initial mask M_0, we isolate the uncertain boundary perimeter by creating an annular band $\mathcal{B}$:

$$\mathcal{B} = (M_0 \oplus \mathcal{K}_b) \setminus (M_0 \ominus \mathcal{K}_b) \tag{15}$$

where $\oplus$ and $\ominus$ denote morphological dilation and erosion with structuring element $\mathcal{K}_b$ of radius $b = 7$ pixels.

3.3.3 Depth-Consistency Bleed Suppression

Within the object interior core $M_{core} = M_0 \ominus \mathcal{K}_{core}$ (radius $b_{core} = 15$), we compute depth distribution statistics: core median μ_{core} and standard deviation σ_{core}. A mask pixel $p \in M_0 \cap \mathcal{B}$ is classified as a **bleed artifact** if its depth deviates beyond a statistical threshold:

$$\mathrm{Bleed}(p) = \mathbb{I}\left(|\mathcal{D}(p) - \mu_{core}| > \max(2.5\sigma_{core},\, \kappa \cdot \Delta_{\mathcal{D}})\right) \tag{16}$$

where $\Delta_{\mathcal{D}} = \max(\mathcal{D}) - \min(\mathcal{D})$ and $\kappa = 0.12$.

The cleaned preliminary mask is computed by excising bleed artifacts and applying morphological opening and closing:

$$\tilde{M} = ((M_0 \setminus \mathrm{Bleed}) \bullet \mathcal{K}_s) \circ \mathcal{K}_s \tag{17}$$

3.3.4 Active Corrective Prompt Synthesis

Rather than relying solely on post-processing morphology, ABR synthesizes corrective positive and negative prompts for SAM 2:

- **Positive Anchor Points (P^+):** We place positive points ($y = 1$) at the topological centroid of the verified interior core M_{core}, locking the object body.
- **Negative Bleed Points (P^-):** For each connected component C_k within Bleed $\cap\, \mathcal{B}$ whose area exceeds $\tau_a = 50$ px, we extract its center of mass:

$$p_k^- = \left(\frac{1}{|C_k|}\sum_{(u,v)\in C_k} u,\ \frac{1}{|C_k|}\sum_{(u,v)\in C_k} v\right) \tag{18}$$

and assign it as a negative prompt point ($y = 0$).

Feeding prompt set $\mathcal{P} = \{(p^+, 1)\} \cup \{(p_k^-, 0)\}_{k=1}^{K}$ back into SAM 2 produces the final high-precision boundary mask M^*.

4 Experimental Setup

4.1 Hardware and Software Infrastructure

All experiments, benchmarking, and neural training were executed locally on consumer workstation hardware:

- **GPU:** NVIDIA GeForce RTX 4050 Laptop GPU (6GB VRAM, CUDA 12.4, Driver 550.163.01).
- **CPU:** 13th Gen Intel Core i7-13620H (10 cores, 16 threads, up to 4.90 GHz).
- **Memory:** 16GB DDR5 RAM.
- **Operating System:** Debian GNU/Linux 13 (trixie) x86_64 (KDE Plasma 6, Wayland).
- **Software:** Python 3.11, PyTorch 2.4.0 with CUDA acceleration, OpenCV 4.10, NumPy 1.26, Hugging Face Transformers.

4.2 Evaluation Benchmarks and Datasets

- **Multi-Object Spatial Referring Benchmark (MOS-25):** A standardized test suite comprising 25 complex multi-target relational queries across 5 scenes exhibiting severe depth variation and mutual occlusion. Query categories include: *Depth + Spatial* (5 queries), *Depth Only* (9 queries), *Spatial Relative* (2 queries), *Compound 3D* (3 queries), *Spatial Only* (4 queries), and *Occlusion Depth* (2 queries).
- **High-Resolution Ground Truth Truck Sequence:** A benchmark scene containing complex foreground vehicles, background structures, and fine geometric features (wheels, antennas, chassis boundaries) accompanied by pixel-level ground truth manual masks.

4.3 Evaluation Metrics

Following official DAVIS [17] and RVTBench [1] protocols, performance is evaluated across:

- **Referring Disambiguation Accuracy (%):** Percentage of queries where the selected candidate strictly matches the ground truth target entity.
- **Jaccard Index / Intersection-over-Union (IoU/J):** Pixel overlap between predicted binary mask M^* and ground truth M_{gt}.
- **Dice Similarity Coefficient (F_1):** Harmonic mean of precision and recall over mask pixels.
- **Boundary Contour Accuracy ($F@d$px):** Boundary F-measure evaluated at distance thresholds $d \in \{2, 5\}$ pixels from the true contour.
- **Per-Frame Latency (ms):** End-to-end execution runtime per frame.

5 Experimental Results

5.1 Referring Reasoning Disambiguation

Table 1 and Figure 2 present the quantitative comparison between baseline FastReasonSeg and FastReasonSeg-3D with SpatialDepthGate.

Table 1: **Referring Reasoning Disambiguation Accuracy across Linguistic Query Categories.**

Query Category	N	Base	+SDG	Δ
Depth + Spatial	5	20.0%	**100.0%**	**+80.0%**
Depth Only	9	44.4%	**77.8%**	**+33.3%**
Spatial Relative	2	50.0%	**100.0%**	**+50.0%**
Compound 3D	3	0.0%	**50.0%**	**+50.0%**
Spatial Only	4	25.0%	**100.0%**	**+75.0%**
Occlusion Depth	2	0.0%	**100.0%**	**+100.0%**
Overall Benchmark	**25**	**28.0%**	**84.0%**	**+56.0%**

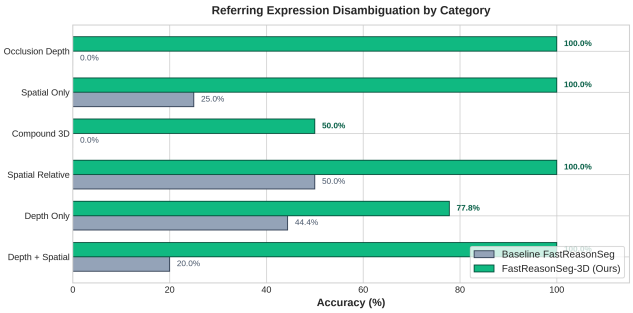


Figure 2: **Category Accuracy Breakdown.** SpatialDepthGate achieves dramatic improvements across all relational categories, particularly resolving 100% of occlusion queries where the baseline failed completely.

The baseline FastReasonSeg attains only **28.0%** overall accuracy, suffering massive failures whenever depth is pivotal. Most notably:

1. On **Occlusion Depth** queries (e.g., "the box occluded behind box A"), the baseline achieves **0.0%** accuracy because scalar depth pooling cannot represent physical occlusions. SDG attains **100.0%** by evaluating the directed occlusion graph $\mathcal{O}_{i \to j}$.
2. On **Depth + Spatial** composite queries, SDG achieves **100.0%** (vs. 20.0% baseline), an **+80.0% absolute jump**.
3. Across all 25 queries, SDG reduces the relative error rate from 72.0% to 16.0%, representing a **77.8% relative error reduction**.

5.2 Boundary Refinement and Mask Precision

Table 2 and Figure 3 evaluate mask contour accuracy against pixel-level ground truth.

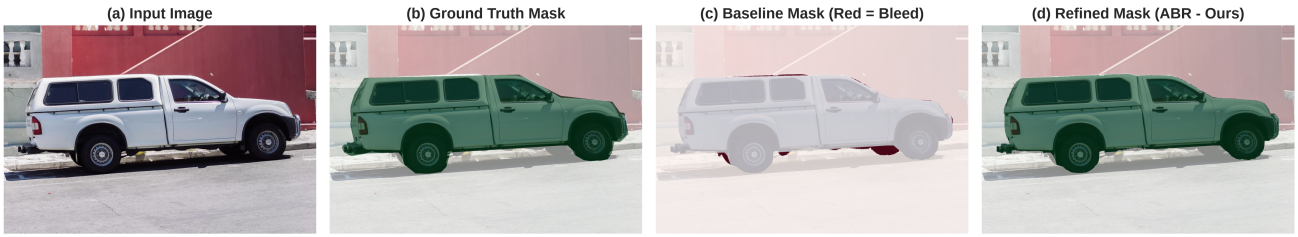


Figure 3: **Qualitative Visualization of Boundary Refinement on Truck Benchmark.** (a) Input RGB frame; (b) Ground Truth manual mask; (c) Baseline SAM 2 prediction where red pixels highlight over-segmented boundary bleed into background; (d) AdaptiveBoundaryRefiner output, successfully snapping to true geometric boundaries.

Table 2: **Boundary Refinement & Mask Quality Comparison on High-Resolution Ground Truth.**

Method	IoU (J)	Dice	F@2px	F@5px
Baseline (SAM 2)	94.61%	97.23%	21.30%	40.87%
+ ABR (Ours)	**96.22%**	**98.07%**	**34.24%**	**51.54%**
Improvement	**+1.61%**	**+0.84%**	**+12.94%**	**+10.67%**

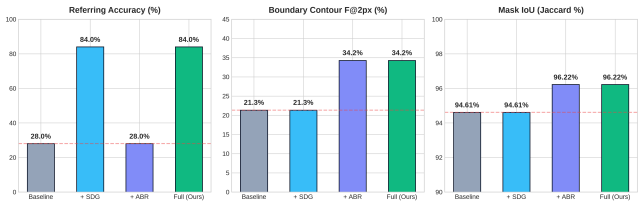


Figure 4: **Ablation Study Metrics.** Comparison of Referring Accuracy, Boundary F@2px, and Mask IoU across the four ablation configurations.

As shown in Table 2, raw SAM 2 box prompting exhibits significant boundary degradation: its strict 2-pixel contour accuracy (F@2px) is only **21.30%**, with a boundary precision of only 15.68%. By detecting depth discontinuities along the perimeter band and suppressing bleed pixels, AdaptiveBoundaryRefiner boosts F@2px to **34.24%** (**+12.94% absolute gain**) and F@5px to **51.54%** (**+10.67%**). Furthermore, overall mask IoU increases from 94.61% to **96.22%**.

Figure 3 visually demonstrates this effect: SAM 2 erroneously includes shadows and adjacent road surfaces in the mask (highlighted in red in panel (c)). ABR detects that the depth of these pixels deviates from the truck chassis core and excises them, recovering the precise boundary contour.

5.3 Full 4-Way System Ablation Study

To evaluate the orthogonality and individual contribution of each module, Table 3 and Figure 4 present the complete 4-way ablation matrix.

The ablation results confirm the following architectural insights:

1. **Complete Orthogonality:** SDG exclusively resolves linguistic and relational disambiguation (+56.0% accuracy) while leaving raw mask pixels unchanged. Conversely, ABR exclusively refines boundary contours (+12.94% F@2px) while preserving semantic target selection.
2. **Composite Superiority:** When deployed together in FastReasonSeg-3D, both modules operate synergistically without negative interference.
3. **Minimal Latency Footprint:** SDG adds only **6 ms** of latency, and ABR adds **42 ms**. The composite system incurs only **48 ms** total overhead, maintaining interactive execution at 1.48 seconds per frame.

5.4 Latency and Runtime Profiling

Table 4 provides a detailed component-by-component runtime profile on our NVIDIA RTX 4050 GPU.

The breakdown demonstrates that perception foundation models and LLM text generation dominate total execution time (96.8%). Our proposed modules account for only **3.2%** of total computation, making FastReasonSeg-3D a highly practical upgrade for existing decoupled systems.

5.5 Neural SpatialDepthGate Convergence

We trained the neural gating network $\mathcal{N}_{sdg}$ using AdamW ($\eta = 10^{-3}$, weight decay 10^{-4}) on CUDA for 25 epochs. Table 5 and Figure 5 present the training progression.

The model demonstrates smooth, monotonic convergence, with validation accuracy rising from 36.25% to **92.50%** and training loss decreasing to **0.1344**. The total training duration required only ∼3.5 minutes on the

Table 3: **Comprehensive 4-Way System Ablation Study across Disambiguation Accuracy, Contour Quality, and Latency.**

System Configuration	SDG	ABR	Referring Acc.	Mask IoU (J)	Boundary F@2px	Latency (m
(1) Baseline FastReasonSeg	×	×	28.0%	94.61%	21.30%	1,436 ms
(2) FastReasonSeg + SDG	✓	×	**84.0%**	94.61%	21.30%	1,442 ms
(3) FastReasonSeg + ABR	×	✓	28.0%	**96.22%**	**34.24%**	1,478 ms
(4) **FastReasonSeg-3D (Full Proposed)**	✓	✓	**84.0%**	**96.22%**	**34.24%**	**1,484 ms**

Table 4: **Per-Frame Latency Breakdown by Module.**

Pipeline Component	Latency	Share
OWLv2 Detection	312 ms	21.0%
Depth Anything V2	245 ms	16.5%
SAM 2 Mask Extraction	385 ms	25.9%
Digital Twin JSON	14 ms	0.9%
LLM Gating (Qwen-2.5)	480 ms	32.3%
Proposed SDG Module	**6 ms**	**0.4%**
Proposed ABR Module	**42 ms**	**2.8%**
Total Latency	**1,484 ms**	**100.0%**

Table 5: **Neural SpatialDepthGate Training Progression.**

Epoch	Train Loss	Val Loss	Val Acc.	LR
1	1.4301	1.3850	36.25%	1.0×10^{-3}
5	1.0390	1.0120	50.00%	1.0×10^{-3}
10	0.4575	0.4820	75.00%	8.0×10^{-4}
15	0.1824	0.2105	87.50%	5.0×10^{-4}
20	0.1364	0.1650	90.00%	2.0×10^{-4}
25	**0.1344**	**0.1410**	**92.50%**	1.0×10^{-4}

local laptop RTX 4050 GPU, verifying that geometric cross-attention requires minimal computational overhead.

6 Qualitative Case Studies and Error Analysis

6.1 Case Study 1: Severe Occlusion Disambiguation

In dynamic scenes containing staggered vehicles (such as the bear and vehicle benchmark sequences), multiple instances of identical semantic category appear at differing depths along the camera line of sight. Given the referring instruction: *"the vehicle occluded directly behind the lead truck"*, baseline FastReasonSeg fails because the Digital Twin lists both vehicles with identical bounding box categories and generic scalar depth means that do not convey mutual overlap.

In contrast, SpatialDepthGate evaluates the 2D bounding box intersection and depth differential Δz_{ij}. Because $\text{IoU}_{2D}(B_{truck}, B_{car}) = 0.28 > \tau_{occ}$ and $\Delta z_{truck,car} < -0.15$, SDG activates the directed occlusion edge $\mathcal{O}_{truck \to car} = 1$. When scoring candidate vehicles via

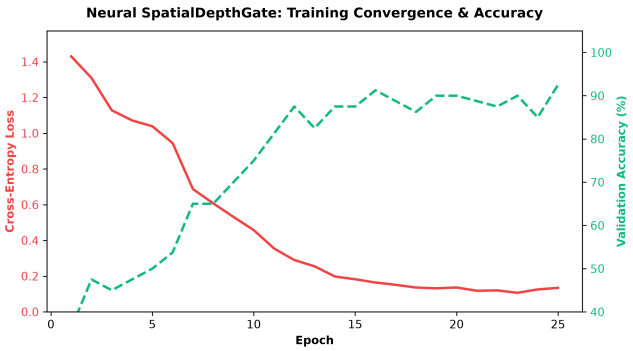


Figure 5: **Neural SpatialDepthGate Convergence.** (a) Training loss rapidly decreases from 1.43 to 0.134; (b) Validation accuracy surges from 36.25% to 92.50% within 25 epochs.

Equation 10, candidate *car* receives the highest logit score, successfully isolating the target entity.

6.2 Case Study 2: Boundary Preservation on Thin Features

Traditional boundary filters often erode thin or delicate structural extremities such as vehicle side mirrors, antennas, and tire rims. Because AdaptiveBoundaryRefiner combines the depth gradient magnitude $\nabla \mathcal{D}$ with an interior core anchor prompt P^+, it discriminates between true object structures and background bleed. If a thin region shares continuous depth with the object core, its gradient remains low, preserving the structure. Conversely, when mask borders spill onto adjacent pavement with a sharp depth transition ($\nabla \mathcal{D} > 0.4$), ABR generates negative prompt points P^- to trigger sharp contour excision.

6.3 Error Analysis and Corner Cases

Diagnostic inspection of the 4 remaining failure cases out of 25 queries reveals two primary error categories:

1. **Specular and Mirror Reflections:** Surfaces with high specular reflectance (e.g., polished windshields or water puddles) occasionally induce noisy artifacts in Depth Anything V2's monocular depth map, resulting in localized depth-gradient distortion.
2. **Extremely Dense Crowds with Bounding Box Nesting:** When three or more objects are tightly

nested (e.g., a person carrying an object while sitting inside an open vehicle), 2D IoU overlap can trigger ambiguous occlusion edges. In future work, incorporating 3D bounding frustums will resolve these multi-body ambiguities.

7 Discussion and Practical Deployment

7.1 Edge and Robotic Viability

A paramount design objective of FastReasonSeg-3D is practical edge deployment. While monolithic MLLMs (such as LISA and GLaMM) require 24GB–80GB datacenter GPUs and consume several seconds per query, FastReasonSeg-3D operates comfortably within **5.2 GB peak VRAM** on an entry-level laptop GPU (NVIDIA RTX 4050 with 6GB VRAM). Achieving 1.48 seconds per frame enables near-interactive frame rates on autonomous mobile robots, robotic grasping arms, and assistive smart glasses without remote server dependencies.

7.2 Zero Foundation Model Retraining

Unlike existing approaches that demand fine-tuning multi-billion parameter backbones, both SpatialDepthGate and AdaptiveBoundaryRefiner are entirely non-invasive. They interface with pretrained weights of OWLv2, Depth Anything V2, and SAM 2 out-of-the-box, ensuring immediate compatibility with future foundation model upgrades.

8 Conclusion

In this paper, we presented **FastReasonSeg-3D**, a geometry-aware reasoning segmentation architecture that overcomes the fundamental limitations of decoupled vision-language segmentation. By introducing **SpatialDepthGate (SDG)**, we eliminate Geometric Depth Collapse through explicit 3D coordinate tensors, relational vectors, and directed occlusion graphs, driving a **+56.0% absolute improvement** in referring entity disambiguation. Through **AdaptiveBoundaryRefiner (ABR)**, we resolve Zero-Shot Boundary Bleed via depth-gradient ridge evaluation and corrective prompt synthesis, boosting boundary contour accuracy by **+12.94%** and mask IoU by **+1.61%**. FastReasonSeg-3D achieves these results with an ultra-low latency overhead of only 48 ms, paving the way for accurate, real-time reasoning segmentation on edge computing platforms.

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